BANKING MANAGEMENT SYSTEM

WEEK 2 – HASHMAP STORAGE

ALGORITHM

OBJECTIVE	To implement Banking Management System using **HashMap<Integer, Account>** where Account ID is used as the key.

1 Start the program.

2 Create a **HashMap<Integer, Account>** named **accounts**.

3 Display the main menu:

1. Create Account
2. Deposit Money
3. Withdraw Money
4. Check Balance
5. Exit

4 Read the user's choice.

5 **If the choice is 1 (Create Account):**

- Read Account ID, name, and initial balance.
- Check if **accounts.containsKey(accountId)** is true.
- If true, display **"Account ID already exists."**
- If false, create a new Account object.
- Store the account in the HashMap using **accounts.put(accountId, account)**.
- Display **"Account created successfully."**

6 **If the choice is 2 (Deposit Money):**

- Read Account ID and deposit amount.
- Use **accounts.get(accountId)** to retrieve the account.
- If account is null, display **"Account not found."**
- If account exists, add the deposit amount to the account balance.
- Display **"Amount deposited successfully."**

7 **If the choice is 3 (Withdraw Money):**

- Read Account ID and withdrawal amount.
- Use **accounts.get(accountId)** to retrieve the account.
- If account is null, display **"Account not found."**
- If withdrawal amount > balance, display **"Insufficient balance."**
- Otherwise, subtract the amount from the balance.
- Display **"Amount withdrawn successfully."**

8 **If the choice is 4 (Check Balance):**

- Read Account ID.
- Use **accounts.get(accountId)** to retrieve the account.
- If account is null, display **"Account not found."**
- If account exists, display Account ID, Account Holder Name, and Balance.

9 **If the choice is 5 (Exit):**

- Display **"Thank you for using Banking Management System."** and terminate the program.

10 For choices 1 to 4, return to the main menu.

11 Stop the program.

KEY POINTS

- Data is stored using **HashMap<Integer, Account>**.
- Account ID is used as the key.
- Direct lookup is performed using **accounts.get(accountId)**.
- Duplicate checking is done using **accounts.containsKey(accountId)**.
- No manual search loops are used for finding an account.
- All data is temporary and will be lost when the program terminates.